



Dissertation on
DeNDDron Swarm - Decentralized Naval Defense Drone Swarm

Submitted in partial fulfillment of the requirements for

Bachelor of Technology
in
Computer Science & Engineering

UE23CS320A – Capstone Project Phase - 1

Submitted by:

Uday Gopan	PES2UG23CS659
Annshul Shetty	PES2UG23CS079
VVS Manohar	PES2UG23CS666
Kasetty Srikar	PES2UG23CS261

Under the guidance of

Dr. Vinodha K

Associate Professor

PES University

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
FACULTY OF ENGINEERING
PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)

Electronic City, Hosur Road, Bengaluru – 560 100, Karnataka, India



PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)

Electronic City, Hosur Road, Bengaluru – 560 100, Karnataka, India

FACULTY OF ENGINEERING

CERTIFICATE

This is to certify that the dissertation entitled

DeNDDron Swarm-Decentralized Naval Defense Drone Swarm'

is a bonafide work carried out by

**Uday Gopan
Annshul Shetty
VVS Manohar
Kasetty Srikar**

**PES2UG23CS659
PES2UG23CS079
PES2UG23CS666
PES2UG23CS261**

In partial fulfillment for the completion of Fifth-semester Capstone Project Phase - 1 (UE23CS320A) in the Program of Study -Bachelor of Technology in Computer Science and Engineering under rules and regulations of PES University, Bengaluru during the period Aug 2025 – Dec. 2025. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The dissertation has been approved as it satisfies the 5th-semester academic requirements in respect of project work.

Signature
Dr. Vinodha K
Professor

Signature
Dr. Sandesh B J
Chairperson

External Viva

Name of the Examiners

Signature with Date

1. _____

2. _____

DECLARATION

We hereby declare that the Capstone Project Phase - 1 entitled “**DeNDDron Swarm - Decentralized Naval Defense Drone Swarm**” has been carried out by us under the guidance of **Dr. Vinodha K, Associate Professor** and submitted in partial fulfillment of the course requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering of PES University, Bengaluru** during the academic semester Aug. – Dec 2025. The matter embodied in this report has not been submitted to any other university or institution for the award of any degree.

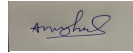
PES2UG23CS659

Uday Gopan



PES2UG23CS079

Annshul Shetty



PES2UG23CS666

VVS Manohar



PES2UG23CS261

Kasetty Srikar



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ABSTRACT

Modern naval vessels face increasing threats from munitions such as rockets, drones and even missiles. Existing defense systems are multi-layered and capable, however they are complex, role specific and expensive. To overcome these issues DenDDron Swarm explores the use of swarms of commodity drones to intercept and neutralize aerial threats which may have been deemed as low value threats that do not require the action of high value defense systems or threats that have slipped past the longer range defense systems. DeNDDron Swarm's intended role is to complement existing defense systems. A decentralized framework is used to control the task assignment and path planning of components of the swarm with tasks being introduced into the swarm via TPA/CPA systems onboard the ship which performs threat classification as well as provide threat vectors for threat interception. Decentralised MAPF algorithms are explored, to be used with a consensus mechanism to share the state of the system among all agents. Task assignment is done based on cost functions derived from careful consideration of metrics to be used for the aforementioned scenario. This unified framework will act as the core software that will run on all agents in the UAV swarm.

TABLE OF CONTENTS

Chapter No.	Title	Page No.
1.	INTRODUCTION	01
2.	PROBLEM STATEMENT	02
3.	LITERATURE SURVEY ‘	03
	3.1 Multi-Agent Pathfinding for Unmanned Aerial Vehicles	03
	3.2 Improved Artificial Potential Field Method Applied for AUV Path Planning	04
	3.3 Comparing the Performance of Zenoh, MQTT, Kafka, and DDS	06
	3.4 Decentralized UAV Swarm Control Architecture	09
4.	MARKET GAP	11
5.	OBJECTIVES	12
6.	ALGORITHM TESTING AND ANALYSIS	14
	6.1 MAPF Algorithms	14
	6.1.1 Artificial Potential Field Algorithm	14
	6.1.2 Optimal Reciprocal Collision Avoidance	20
7.	CONCLUSION OF CAPSTONE PROJECT PHASE - 1	22
8.	PLAN OF WORK FOR CAPSTONE PROJECT PHASE - 2	23
	REFERENCES	24
	APPENDIX A DEFINITIONS, ACRONYMS, AND ABBREVIATIONS	25

LIST OF TABLES

Table No.	Title	Page No.
1	Framework latency comparison table for single and multi node	8
2	Metrics for 10 Agents while using APF algorithm	17
3	Metrics for 100 Agents while using APF algorithm	18
4	Metrics for 1000 Agents while using APF algorithm	19

LIST OF FIGURES

Figure No.	Title	Page No.
1.	GNRON fix using distance correction	5
2.	Local minima fix using virtual regular hexagon	5
3.	Dynamic collision avoidance using relative velocity	5
4.	Framework throughput comparison architecture	7
5.	Framework latency comparison architecture	7
6.	Throughput data in msgs/s for multi-node scenario	7
7.	Throughput data in bits/s for the multiple-node scenario	8
8.	Curve used by APF in MAPF	16
9.	APF Simulation 1	16
10.	APF Simulation 2	17
11.	APF Simulation 3	17
12.	10 Agent APF Test	17
13.	100 Agent APF Test	18
14.	1000 Agent APF Test	19
15.	ORCA Permissible Velocities	20
16.	ORCA working diagram	21
17.	ORCA Simulation 1	22
18.	ORCA Simulation 2	22

CHAPTER 1

INTRODUCTION

The most important task when dealing with dynamic and hostile environments that the naval forces have to deal with on an everyday basis, this is where job assignment and multi-agent path planning become of the utmost priority. The systems that have been assigned to deal with threats like the proposed decentralized swarm drone system would have to assign tasks to each node and design collision free path which intercept the threats, our goal is to use these decentralized systems as our naval forces would often have to work in gps denied environments with high communication latency.

The existing systems are centralized systems that have to deal with issue of single point of failure and need an extremely high communication bandwidth which becomes a major problem in the environments where naval forces usually have to intercept and neutralize these threats. Our goal is to directly address this problem by the evaluation and simulation of multiple decentralized multi agent path finding strategies using a 3D simulation framework, we have included both single and multi agent path finding algorithms for example A* (A star) algorithm, RRT* (Rapidly-exploring Random Tree), reactive APF (Artificial Potential Field) and cooperative APF strategy. These algorithms will be simulated under realistic simulations of naval requirements.

Our aim is to show that a decentralized network of drones communicating using a peer to peer framework would perform better for threat interception in conditions where traditional systems would be insufficient.

CHAPTER 2

Problem Statement

In the current scenario where there is an increasing number of threats in naval warfare, like anti-ship missiles, and coordinated swarm attacks. All of these pose a significant challenge to the defense systems. While there are already defense mechanisms to intercept these threats like SAMs (Surface-to-Air Missiles) and CIWS (Close-In Weapon Systems), they are constrained by high costs, limited ammunition and not very reliable in some situations which can cause the defense to be compromised. Moreover, these systems heavily rely on centralized system which means there exists a single server which controls all of them, this is very ineffective because this can lead to single point of failure and also in GPS denied environments these systems fail to communicate leaving them exposed to adversaries.

Existing defense mechanisms also struggle with scalability. Even though a centralized approach enables precise coordination because it's being controlled by one server, it lacks the resilience to maintain effectiveness in critical situations. This inability to adapt to the situation and failing to redistribute tasks among available agents results in gaps that can be exploited causing loss to the vessels.

There is a dire need for solutions that combine the coordination from decentralized algorithms with robust multi-agent path finding techniques which enables autonomous drone swarms to operate as an aerial defense layer. DeNDDron swarm proposes a decentralized swarm framework that unifies MAPF and task assignment to coordinate multiple commodity UAV units. Built on top of a robust consensus mechanism and relative localization assumptions.

With this approach the framework is designed to enhance defensive capabilities while also reducing dependence on expensive single use ammunition. This multi-layered approach aims to fill existing gaps in adaptive threat response, while also establishing cost-effective, scalable foundations for next-generation defense systems that can operate in the most challenging combat environments.

CHAPTER 3

Literature Survey

3.1 Multi-Agent Pathfinding for Unmanned Aerial Vehicles

(Kymry Burwell)[1]

3.1.1 Introduction

Kymry Burwell, in this paper addresses the challenge of having to coordinate multiple UAVs in real time using (MAPF) algorithms. His research recognises that the MAPF algorithms have been studied in 2D environments, but operations in 3D airspace present an unique challenge with multiple dynamic constraints.

3.1.2 Methodology

This paper goes through various state of the art MAPF algorithms some of them being, CBS (Conflict-Based Search), A*, OD (Operator Decomposition) etc... and provides an extensive comparative analysis of all of them.

3.1.3 Key findings

Online MAPF methods enable dynamic replanning but sacrifice global optimality.

3D airspace reduces obstacle constraints but introduces altitude coordination challenges.

The paper identifies several gaps that remain unaddressed for UAV swarm operations, some of them being communication constraints, decentralization, dynamic threats, etc..

3.1.4 Evaluation

Overall the paper identifies several critical gaps that remain unaddressed for UAV swarm operations in real-world scenarios.

3.2 Improved Artificial Potential Field Method Applied for AUV Path Planning [2]

3.2.1 Introduction

This paper explores the use of artificial potential field path planning for AUV path planning. It touches upon the fitting of APF in unknown scenarios where real time path planning and correction is required. It adapts well to dynamic environments that have both static as well as dynamic obstacles. The paper discusses an improved APF algorithm that addresses real time path planning while overcoming hurdles like GNRON and local minima by construction of virtual regular hexagons as well as evolutionary potential fields for dynamic obstacle avoidance.

3.2.2 Methodology

The paper uses standard attractive field and the following repulsive field:

$$U_{\text{rep}}(X) = \begin{cases} \frac{1}{2}k_{\text{rep}} \left(\frac{1}{d(X, X_o)} - \frac{1}{d_0} \right)^2, & d(X, X_o) \leq d_0, \\ 0, & d(X, X_o) > d_0, \end{cases} \quad (1)$$

Forces for each are found by differentiating the field to find the gradient pertaining to the location of the agent. Resolution of the forces found from summation attraction and repulsion gives the final force. The solution the authors have come up with to tackle to the GNRON issue is to apply a distance correction factor to the repulsive piecewise function as shown in Fig. 1

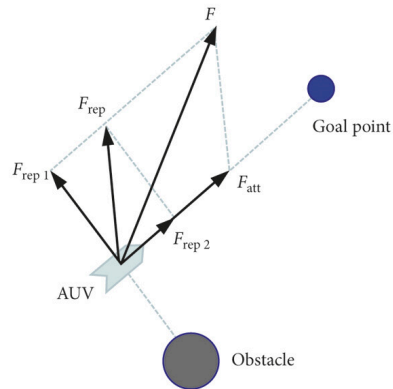


Fig. 1: GNRON fix using distance correction

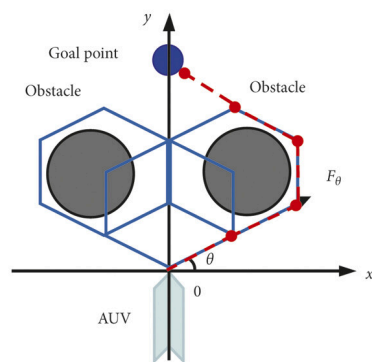


Fig. 2: Local minima fix using virtual regular hexagon

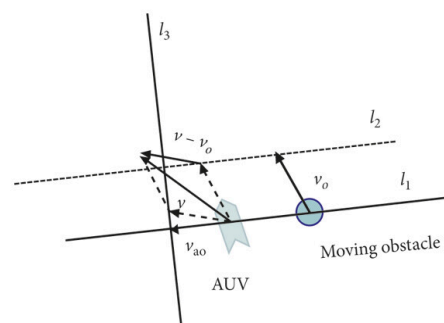


Fig. 3: Dynamic collision avoidance using relative velocity

The second improvement made by the authors is with respect to the local minima problem, this was done by applying a virtual regular hexagon path around the obstacles. This is illustrated in Fig 2.

The final improvement made is with regards to the dynamic collision avoidance, this is done by considering the direction and magnitude of velocity alongside the spatial location, the paper proposes a relative velocity method to achieve this, the improved force function is given by:

$$F_{\text{rep}}(X, V) = -\nabla U_{\text{rep}}(X, V) = \begin{cases} F(X) + F(V), & d(X, X_o) \leq d_0 \& v_{ao} \geq 0, \\ 0, & \text{else,} \end{cases} \quad (5)$$

This relative correction is demonstrated in Fig 3.

3.2.3 Key Findings

The proposed methods are able to successfully resolve path planning problems of both static as well as dynamic obstacles. The solution gracefully handles GNRON, local minima as well as dynamic collision scenarios in real time.

3.2.4 Evaluation

This paper addresses some highly relevant problems that plague APF algorithms gracefully such as distance correction, virtual regular hexagons and relative velocity based corrections.

3.3 Comparing the Performance of Zenoh, MQTT, Kafka, and DDS [3]

3.3.1 Introduction

This article compares the performance of various frameworks like Zenoh, MQTT, Kafka and Cyclone DDS on single system as well as multi-node systems. It compares two metrics, the throughput and latency on a single system and ethernet connected network.

3.3.2 Methodology

Fig. 4 and Fig. 5 show the architecture that has been employed to compare the throughput and latencies of the various frameworks. Testbed configuration: Ubuntu 20.04.3 LTS with a 4GHz 8 core

16 thread processor having 32GB DDR4 memory and a 100Gb Ethernet connection. The throughput comparative study was conducted in both single machine as well as multi machine scenarios on CycloneDDS, Zenoh P2P, Zenoh Brokered, MQTT and Kafka. Only multiple machine scenarios are explored in this report given its relevance to the project. Following are the throughput comparison graphs.

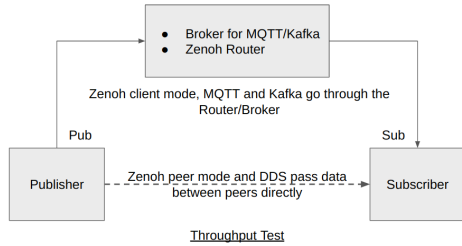


Fig. 4: Framework throughput comparison architecture

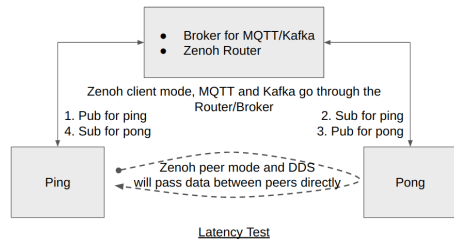


Fig. 5: Framework latency comparison architecture

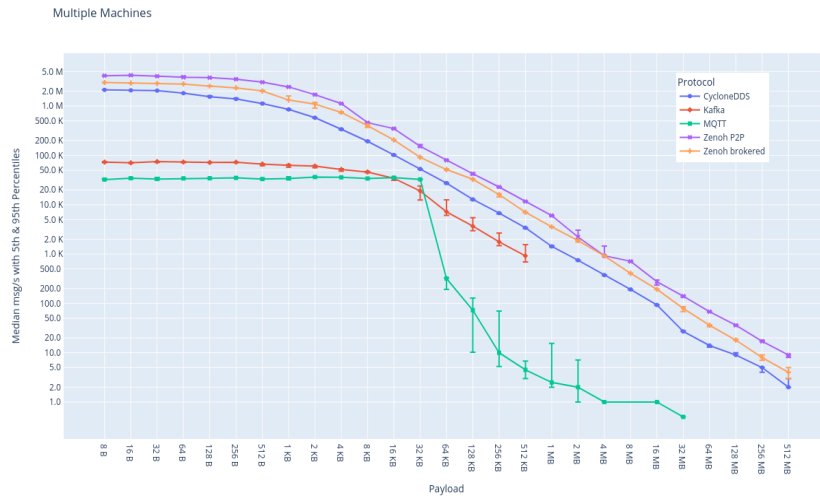


Fig. 6: Throughput data in msgs/s for multi-node scenario

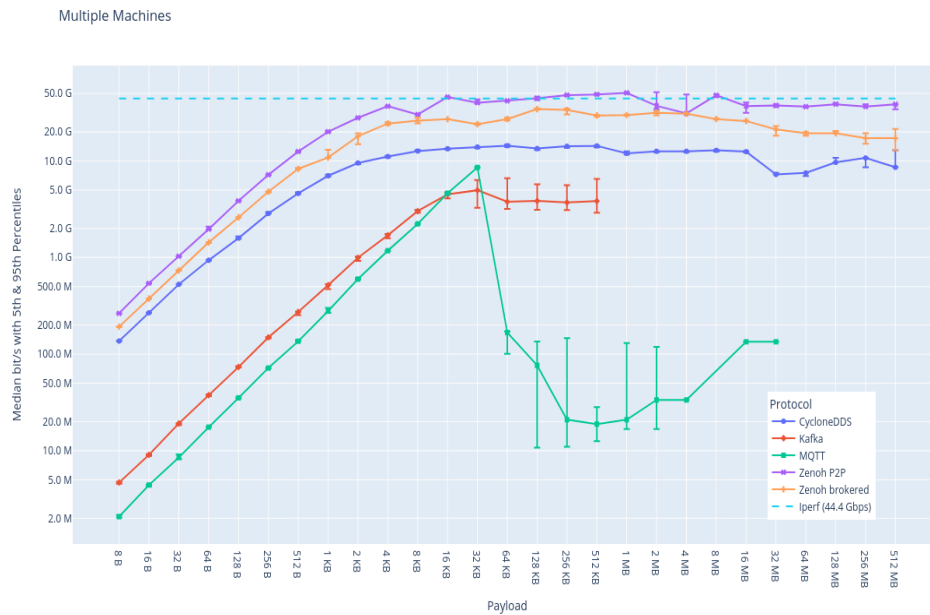


Fig. 7: Throughput data in bits/s for the multiple-node scenario

The latency comparative study was conducted in single machine as well as multi-node configurations using the frameworks Kafka, MQTT, CycloneDDS, Zenoh brokered, Zenoh P2P, Zenoh-pico and the

ping tool. Table 1 contains the latency measured using the 1st and 99th percentile trimmed data with the result taken as half of the median RTT. The following integers are of the unit *microsecond*.

Target	Single-machine	Multiple-machine
Kafka	73	81
MQTT	27	45
Cyclone DDS	8	37
Zenoh brokered	21	41
Zenoh P2P	10	16
Zenoh-pico	5	13
ping	1	7

Table 1: Framework latency comparison table for single and multi node

3.3.3 Key Findings

From the results that have been found by experimentation by the Taiwanese research team, we can conclude that Zenoh has the best performance in terms of both throughput as well as latency. The brokered Zenoh and P2P Zenoh appear to perform worse than CycloneDDS because of the lack of

multicast functionality, however the Zenoh-pico made for low resource systems has bridged this gap and gotten near instantaneous communication.

3.3.4 Evaluation

Further studies on the Zenoh framework shows very promising prospects. The low resource framework, Zenoh-pico shows excellent performance with 13microsecond median communication latency and a 25M msgs/s throughput for 8Kb messages. This has narrowed the scope of frameworks to choose between DDS models like CycloneDDS and the more promising Zenoh.

3.4 Decentralized UAV Swarm Control Architecture (Xia et al.) [4]

3.4.1 Introduction

The present article outlines a multi-layered decentralized control system for UAV swarms designed to intercept multiple dynamic targets in complex environments. Centralized UAV control systems can become limited in poor environments where communication can become an issue. In this situation a UAV co-ordination systems performance can degrade, and relatively speaking communication can be poor with low fault tolerance to a single failure, and these would be undesirable for an interception that would occur quickly and in real-time.

3.4.2 Methodology

The architecture proposes a multi-layer control mechanism, in which each layer is responsible for its respective operational domain :

- Perception Layer: Responsible for sensing, intruder detection, and state estimation.
- Decision Layer: Based on a hybrid of learning-based models and traditional, rule-based logic to make target assignment and maneuver decisions.

- Control Layer: Execute low-level flight control based on stable, deterministic rules.
- Emergency Mode Layer: Manages the automatic switch to Return-to-Base Mode when faults or critical states occur.

The decentralized nature relies on local communication where representatives -in this case, UAVs- only exchange necessary state information such as target vectors and agent state (flying, walked, landed, etc.). Instead of using a central controller, UAVs evaluate within themselves whether they are a suitable agent to intercept an assigned target employing an internal cost evaluation and subsequently communicate with nearby UAVs to avoid duplication or conflict. Simulations were performed toward multiple scenarios including: multi-intruder interception; UAV failure while executing mission; and emergency mode activation while conducting path planning. These conditions were employed to assess verification regarding task allocation as well as coordination effectiveness and ability to reorganize swarm capability when agents are lost.

3.4.3 Key Findings

The proposed system was found to: Increase target-tracking accuracy because of the hybrid learning plus rule-based approach. Improve task scheduling, because of decentralized decisions processing which reduced time delays in centralized systems. Maintain a high interception success rate as the number of dynamic intruders increased. Exhibit fault tolerance as the system redistributed tasks automatically following the loss of an individual UAV. Support autonomous transitions to a safe-state using the dual flight-mode structure

3.4.4 Evaluation

These findings from the simulation experiments indicated that the proposed decentralized architecture greatly improved Tracking Accuracy and Interception Success, while task scheduling

efficiency were improved over traditional methods. The frame also indicated good scalability to larger swarm sizes as well as maintained performance with UAV failures. But authors recognized the need for testing, and improved transitions of control from the human operator would be necessary to fully deploy.

CHAPTER 4

Market Gap

Naval vessels are some of the highest value assets that are deployed by any military, costing billions to deploy and maintain while also carrying hundreds of sailors and officers. The protection of these are paramount for the security of the nation. Amidst rising tensions and growing militaries, the threat of aerial weapon systems has continued to grow at an alarming pace. These threats ignore the terrain beneath, be it land, water or the air itself. Rising threats bring cheap, disposable munitions such as kamikaze drones and other cheap rocketry. Defense against intelligent systems like drones require the use of advanced defence systems like lasers or EW suites. Using highly expensive systems like SAMs is not viable when there is a large volume of adversaries and systems like CIWS do not guarantee kills. Existing systems have multiple layers of defense, SAMs for medium and long range aerial threat neutralization, CIWS for close range slow moving threats, chaff to throw away radar based fire and forget munitions, laser systems to neutralize small threats. EW suites to jam electronic systems and many more creating a high tech onion which an attacker must get through to reach the ship. DeNDDron Swarm intends to fill in the small gaps that exists in this already advanced armour. To provide a cheap disposable and easily deployable defense grid that can intercept and neutralize any threat within the operational theatre of the UAV used.

To fight fire with fire is the inspiration behind DeNDDron Swarm. To combat cheap commodity hardware using cheap commodity hardware, To create an excellent software framework to protect a naval vessel and its domain from a large variety of threats, ranging from drones to rockets and even missiles. The building of a defense grid framework for moving bodies such as ships

and giving them autonomy for independent motion and decision making forms a robust system which is capable of working in any environment to serve and protect in any scenario.

In conclusion, the framework does not intend to replace any existing system but to strengthen them by adding a flexible defense layer to catch any adversaries that might have slipped the higher levels or to intercept low value, slow moving threats without wasting precious munitions.

CHAPTER 5

Objectives

The DeNDDron Swarm Project has the goal of eliminating the command and control based vulnerabilities of centralized defense of naval vessels by developing a swarm of autonomous drones to patrol an aerial perimeter. The proposed model allows drones, while communicating with other drones, to collaboratively engage threats to the vessels under attack by independently intercepting in response to context cues and will not rely on any one drone (centralized command) to direct engagement of threats or unauthorized access to a given perimeter of a naval vessel. This capacity to coordinate an engaged response among the autonomous drones will still be possible even in the likely event of GPS jamming or loss of communication between drones, and is especially common in maritime combat.

A main function and goal for the swarming drones is to be autonomous and safe in flight while utilizing advanced MAPF (Multi-Agent Path Finding) as well as decentralized algorithms such as A* or RRT*. This will allow the drones to efficiently surround and intercept quickly moving threats without colliding with each other and without the global knowledge of the drones being limited to their location relative to nearby drones.

Another important goal will be to provide a fully autonomous ability to select a threat to engage. There will not be a central assignment of threats, but instead each drone will independently assess the

cost function of distance, energy consumed, or interception time to engage a threat. This makes the swarming drones very adaptable, without the need to wait for a command from a human or drone as the mission can be immediately redistributed in the event of drone failure or responds to a new threat, there will be no need to wait for a centralized command.

Finally, the project will develop a robust and fault-tolerant software system on every drone in order to provide a peer-to-peer communications system together with a shared state and distribute decision making. The new system will be tested for validity in a realistic 3D simulation with ship motion, and sensor delay.

CHAPTER 6

Algorithm Testing and Analysis

6.1 MAPF Algorithms

Two MAPF algorithms have been implemented to test the feasibility of these highly promising algorithms. These algorithms have been selected after rigorous consideration which included parameters like decentralizability, flexibility, completeness, adaptability, fitting with 3D scenarios, potential communications load, computational time and space efficiency, efficiency in sparse environments. Based on these parameters the two MAPF solvers that we have implemented are APF and ORCA. One that uses the mathematical foundations of force gradients created by artificial potential fields as a function of its position and the other that co-operatively solves MAPF problems by splitting the responsibility of collision avoidance and gracefully solving collisions by adjusting own velocity based on the velocities of self and possible colliding agent.

6.1.1 Artificial Potential Field Algorithm

APF algorithm is a well established algorithm that has been used in theoretical studies for SAPF as well as MAPF, with little adoption in real world systems, however recent works[2] have demonstrated the efficacy of the algorithm in real world scenarios when some fundamental issues

like local minima problem, and GNRON are solved. Improving upon and using inspiration from previous works[2] several modifications have been made to the standard APF strategy.

Direct force calculations are applied in the source instead of differentiating the field function to find force during operations. The force function is given by the equations:

Standard unit-direction attractive force (towards goal or active waypoint $\mathbf{x}_{\text{target}}$):

$$\mathbf{F}_{\text{att}}(\mathbf{x}) = \begin{cases} k \frac{\mathbf{x}_{\text{target}} - \mathbf{x}}{\|\mathbf{x}_{\text{target}} - \mathbf{x}\|}, & \mathbf{x} \neq \mathbf{x}_{\text{target}} \\ \mathbf{0}, & \text{otherwise.} \end{cases} \quad (6)$$

Force-ramping when stuck (for ticks $t_{\text{stuck}} > T_{\text{thresh}}$):

$$k(t) = k_0 \exp(\alpha(t_{\text{stuck}} - T_{\text{thresh}})), \quad (7)$$

(8)

Repulsive Force from Static Obstacles

For obstacle i with closest surface point $\mathbf{c}_i$ and distance d_i (considered only if $0 < d_i \leq R_{\text{infl}}$):

$$\mathbf{F}_{\text{rep},i}(\mathbf{x}) = \begin{cases} \frac{1}{b d_i^3} e^{-ad_i} (\mathbf{x} - \mathbf{c}_i), & 0 < d_i \leq R_{\text{infl}}, \\ \mathbf{0}, & \text{otherwise.} \end{cases}$$

(Equivalent to a scalar magnitude $\frac{1}{b d_i^2} e^{-ad_i}$ times the unit vector pointing away from the obstacle.)

Repulsive Force from Other Agents

For another agent at $\mathbf{x}_j$ with inter-agent distance d_j and agent-specific gain K_{agent} (active when $0 < d_j \leq R_{\text{agent_infl}}$):

$$\mathbf{F}_{\text{agent},j}(\mathbf{x}) = \begin{cases} K_{\text{agent}} \frac{1}{b d_j^3} e^{-ad_j} (\mathbf{x} - \mathbf{x}_j), & 0 < d_j \leq R_{\text{agent_infl}}, \\ \mathbf{0}, & \text{otherwise.} \end{cases} \quad (9)$$

Total Force

Sum of attractive and all repulsive contributions:

$$\mathbf{F}_{\text{total}}(\mathbf{x}) = \mathbf{F}_{\text{att}}(\mathbf{x}) + \sum_i \mathbf{F}_{\text{rep},i}(\mathbf{x}) + \sum_j \mathbf{F}_{\text{agent},j}(\mathbf{x}). \quad (10)$$

Hexagon Local-Minima Escape

When the agent is detected as stuck (movement $< \varepsilon$ for T_{thresh} ticks), construct a 2D regular hexagon in the agent's horizontal plane:

Vertices (for $i = 0, \dots, 5$):

$$\mathbf{v}_i = \mathbf{x} + R \begin{bmatrix} \cos \theta_i \\ \sin \theta_i \\ 0 \end{bmatrix}, \quad \theta_i = i \cdot \frac{\pi}{3}. \quad (11)$$

Typical choice: $R = 1.5 R_{\text{infl}}$

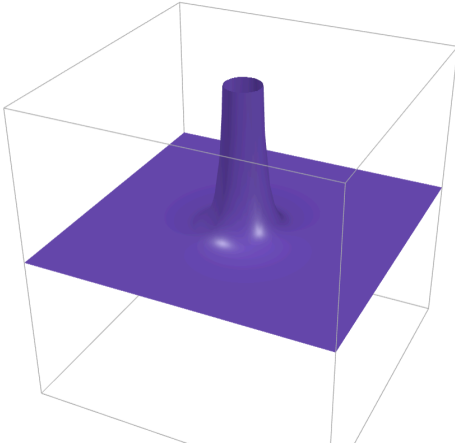


Fig. 8: Curve used by APF in MAPF

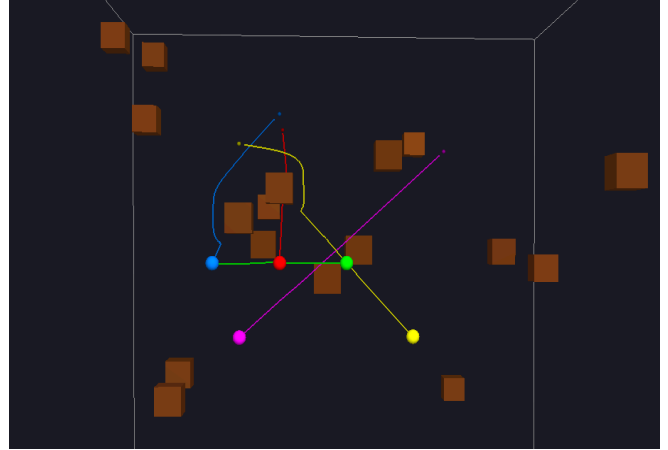


Fig. 9: APF Simulation 1

These equations define the behaviour of agents in all scenarios, inspired by previous work[2], the virtual regular hexagon method has been employed to tackle local minima issue. The novelty in this system lies in its combination of exponential ramping to overcome GNRON alongside virtual regular hexagon method as well as the application of a continuous potential field that is not reliant on the

piecewise function that the standard approaches use. The field equation that has been used in the DeNDDron Swarm implementation of APF produces a curve of the nature as shown in Fig. 8

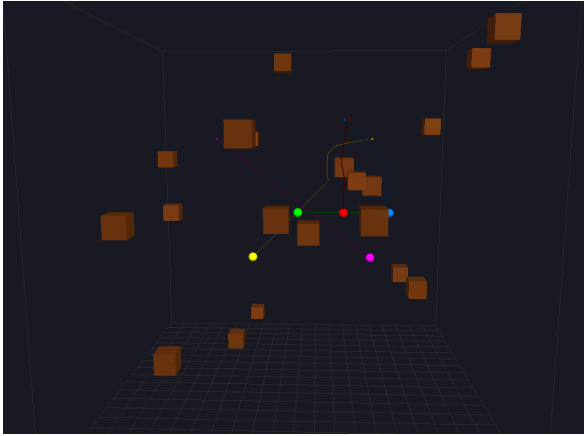


Fig. 10: APF Simulation 2

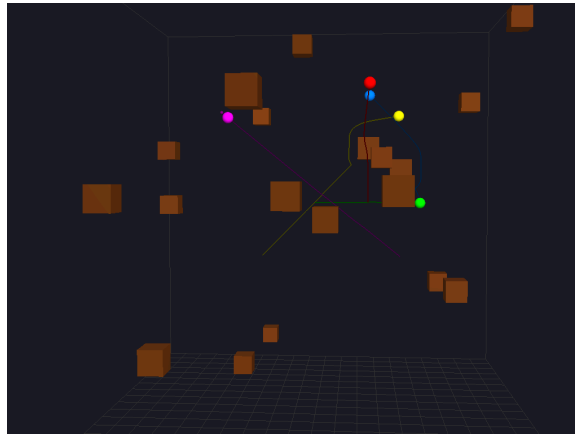


Fig. 11: APF Simulation 3

Fig. 9, Fig. 10 and Fig. 11 illustrate the operation of the APF algorithm in a custom simulation environment that has primarily been created using the C++ library OpenGL. The environment has been designed to be able to take any algorithm using the algorithm interfaces that have been provided in the main multi agent path planner class. This particular experiment was conducted using 20 obstacles placed in pseudo random locations, where the first 4 are placed in random locations and the remaining are placed randomly. 4 agents have been spawned in predetermined locations with predetermined goals. This is not a lifelong MAPF simulation only arrival at destination is expected. The following shows the metrics that have been calculated for the simulated run:

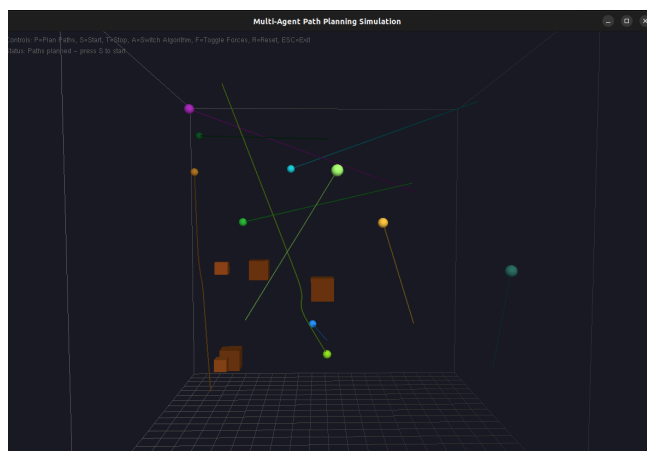


Fig. 12: 10 Agent APF Test

=== Calculating Metrics ===

MAPF METRICS REPORT	
Algorithm: APF-MAPF	
Agents: 10	Obstacles: 5
ESSENTIAL METRICS	
Makespan:	16.881 units
Sum of Costs:	238.332 units
Avg Path Length:	23.833 units
Planning Time:	0.001547 seconds
SUCCESS METRICS	
Agents Reached Goal: 10 / 10 (100.0%)	
QUALITY METRICS	
Avg Efficiency:	100.00% (Min: 100.00% Max: 100.00%)
Direction Changes:	0 total
COLLISION METRICS	
Collision-Free:	YES
Agent Collisions:	0
Obstacle Collisions:	0
Near Misses:	0

Table 2 : Metrics for 10 Agents while using APF algorithm



Fig. 13: 100 Agent APF Test

All agents reached their goals!

=== Calculating Metrics ===

MAPF METRICS REPORT			
Algorithm: APF-MAPF			
Agents: 100	Obstacles: 5		
ESSENTIAL METRICS			
Makespan:	21.597	units	
Sum of Costs:	2366.974	units	
Avg Path Length:	23.670	units	
Planning Time:	0.013435	seconds	
SUCCESS METRICS			
Agents Reached Goal: 100 / 100 (100.0%)			
QUALITY METRICS			
Avg Efficiency:	99.96 % (Min: 97.75 % Max: 100.00%)		
Direction Changes:	0	total	
COLLISION METRICS			
Collision-Free:	YES		
Agent Collisions:	0		
Obstacle Collisions:	0		
Near Misses:	0		

Table 3: Metrics for 100 Agents while using APF algorithm

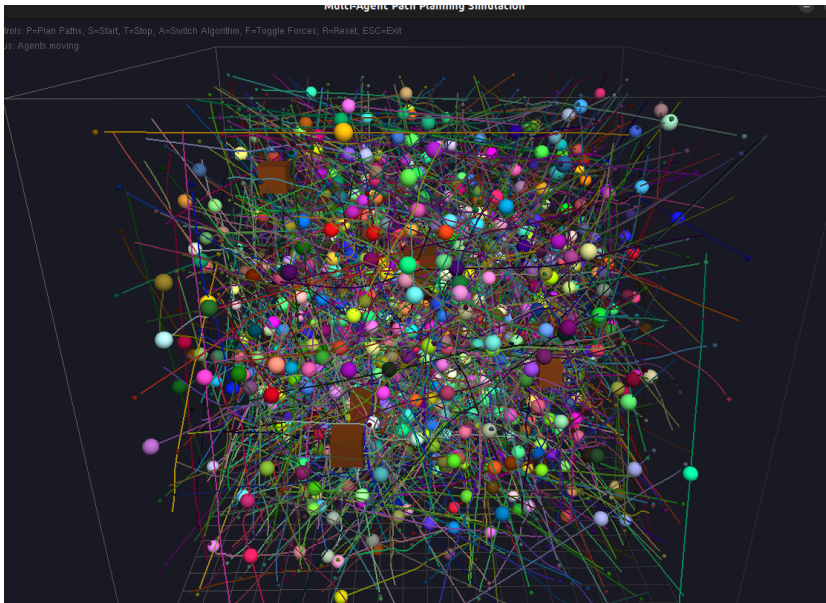


Fig. 14: 1000 Agent APF Test

MAPF METRICS REPORT		
Algorithm: APF-MAPF		
Agents: 1000 Obstacles: 5		
ESSENTIAL METRICS		
Makespan:	114.381	units
Sum of Costs:	25794.631	units
Avg Path Length:	25.795	units
Planning Time:	0.461733	seconds
SUCCESS METRICS		
Agents Reached Goal: 1000 / 1000 (100.0%)		
QUALITY METRICS		
Avg Efficiency:	97.54 % (Min: 10.55 % Max: 100.00%)	
Direction Changes:	1723	total
COLLISION METRICS		
Collision-Free:	YES	
Agent Collisions:	0	
Obstacle Collisions:	0	
Near Misses:	0	

Table 4: Metrics for 1000 Agents while using APF algorithm

6.1.2 Optimal Reciprocal Collision Avoidance

ORCA is a decentralized algorithm where each agent independently calculates its velocity to avoid collisions with others by minimizing deviation from its preferred path.

Selecting the optimal action for each agent comes down to solving a low-dimensional linear program, which drastically reduces the computation cost and therefore having the path as smooth as possible.

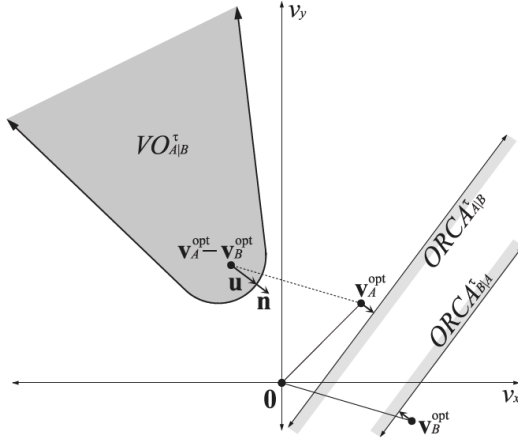


Fig. 15: ORCA Permissible Velocities

To “share the responsibility” of avoiding collisions among the robots in a fair way, robot A adapts its velocity by (at least) $\frac{1}{2}(\mathbf{u})$ and assumes that B takes care of the other half. Hence, the set $ORCA_{A|B}^r$ of permitted velocities for A is the half-plane pointing in the direction of $\mathbf{n}$ starting at the point $\mathbf{v}^{opt}$ formally:

$$ORCA_{A|B}^r = \{\mathbf{v} \mid (\mathbf{v} - (\mathbf{v}_A^{opt} + \frac{1}{2}\mathbf{u})) \cdot \mathbf{n} \geq 0\}.$$

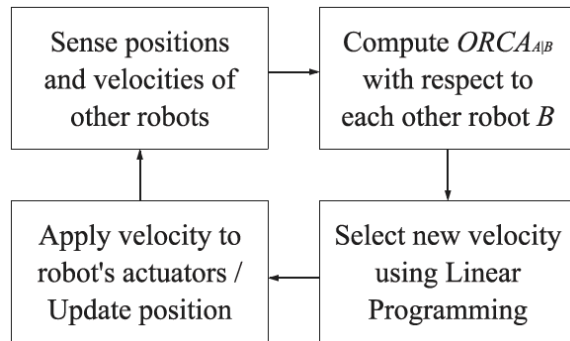


Fig 16.: ORCA working diagram

This is the schematic overview of the continuous cycle of sensing and acting that is independently being executed by each agent.

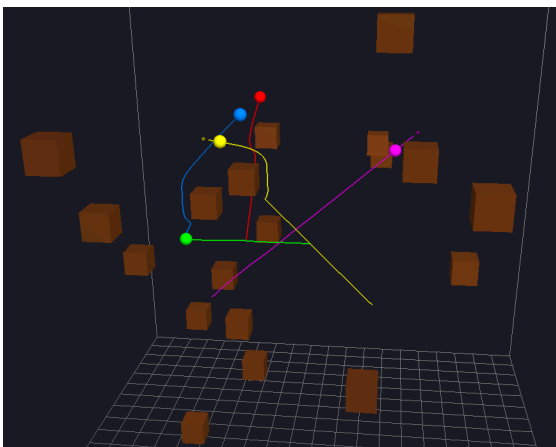


Fig. 17: ORCA Simulation 1

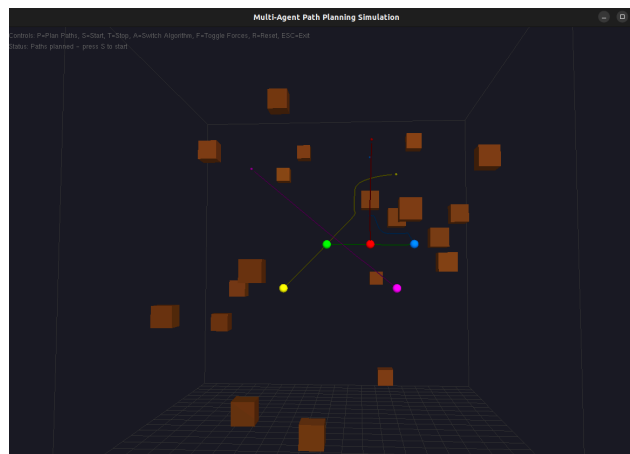


Fig. 18: : ORCA Simulation 2

The figures, 14 and 15 demonstrate the path plotted using ORCA however this algorithm is still in its early stages and requires more testing to provide accurate results.

CHAPTER 7

Conclusion of Capstone Project Phase-1

We have now formed a foundation for our project DeNDDron Swarm, which is a swarm drone algorithm that works on peer to peer communication between the nodes which eliminates the problem of a single point of failure which existing systems have and can become a huge barrier for them to overcome while dealing with threat interception. Currently we have ran dynamic 3D simulations for both single agent and multi agent path finding algorithms for example A*(A star) algorithm, RRT*(Rapidly exploring Random Tree) algorithm, APF (Artificial Potential Field) algorithm. We have learned that from the evaluation and conclusion of these algorithms we have to model a strategy pattern which has a very strong path planning logic that is backing it as highlighted by Kymry Burwell in [3].

After analyzing the market we have noticed there is an absence of decentralized systems in defence which can become a necessity in GPS-denied environments and regions of low bandwidth and with the help of Zenoh, MQTT, Kafka and DDS [3] selection of lightweight communication frameworks

Our goal for the future would be to integrate these algorithms into an optimized path finding strategy, analyze the gap in the market, make use of the research papers we have inferred from build an strong optimization collision-free lightweight framework for communication and path finding which will lead us to build. We want to build the next generation of defense systems forces with our decentralized swarm drone system which will be the breakthrough in naval defense practices in the future.

CHAPTER 8

Plan of Work for Capstone Project Phase-2

With preliminary testing of SAPF and MAPF algorithms being completed along with research on networking technologies for agentic systems, the focus in the second phase will lean towards finalization of the message passing framework, the development of a robust consensus mechanism

and the evolution of the agent singleton. The agent singleton will be a single class with various helpers to perform different actions like path planning, job assignment, consensus etc.. This singleton will be packaged into a docker container for quick deployment and testing scalability as well as the robustness of the consensus mechanism under various simulated network scenarios. Phase 2 will also see the introduction of the controller node, which performs the role of metric generation and introduction of jobs to the swarm. Future work will include visualization or numerical simulation of the working of all systems integrated into a unified testbed to perform both independent as well as integrated testing. Integration of path finding algorithms with the distributed frameworks is expected to be a major hurdle as distributed system trade-offs wrt CAP theorem become a major concern. The knowledge of the state of the system at every tick requires a highly consistent yet available network where partition tolerance is non-negotiable. Implementation of the aforementioned modules and classes is the goal of Phase 2 which is expected to bleed into Phase 3 alongside literature drafting.

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Appendix A: Definitions, Acronyms and Abbreviations

Term / Acronym	Definition
A*	A-star ; an optimal graph based shortest-path algorithm using heuristics.
APF	Artificial Potential Field;a path planning method based on attracting and repelling forces.
AUV	Autonomous Underwater Vehicle; mentioned in the literature as a potential improvement to APF.
CAP Theorem	States that a distributed system cannot guarantee Consistency, Availability, and Partition Tolerance simultaneously.

CBS	Conflict-Based Search; an optimal MAPF approach that uses a hierarchical way to identify conflict.
CIWS	Close-In Weapon System; A short-range naval defense system.
CPA / TPA	Closest Point of Approach / Time to Point of Approach; metrics tracking the interception of a threat.
CycloneDDS	Middleware that uses DDS both for real-time and decentralized communication.
DDS	Data Distribution Service; a distributed communications protocol used in robotics.
DoS	Denial of Service; one type of disruption attack that negatively impacts communications or computation resources.
EW	Electronic Warfare; refers to equipment that jams or spoofs enemy electronics.
GNRON	Goal Not Reachable with Nearby Obstacles; an APF failure mode wherein from a repelling force, a goal cannot be reached.
Kafka	Distributed message-streaming middleware with high-throughput.
MAPF	Multi-Agent Path Finding; algorithms producing collision-free paths for multiple agents.

MQTT	Lightweight publish–subscribe protocol suitable for constrained devices and constrained networks.
OD	Operator Decomposition; a MAPF algorithm that breaks down problems into operations for use in a particular framework.
ORCA	Optimal Reciprocal Collision Avoidance; a decentralized velocity collision-avoidance algorithm.
PRM	Probabilistic Roadmap Method; a probability sampling-path planning method.
RRT / RRT*	Rapidly-Exploring Random Tree (or the optimal variant RRT*); a family of sampling-based motion-planning algorithms.
SAM	Surface-to-Air Missile; A long-range threat-intercepting system, which is used to intercept aerial threats.
SAPF	Single-Agent Potential Field; An APF that was used to plan the path of a single agent.
Swarm Consensus	A distributed decision-making mechanism that is used to synchronize all the state of agents.
Throughput	The messages or bytes transmitted per second as part of a test of a communication performance.
UAV	Unmanned Aerial Vehicle; A drone used as a mobile agent.